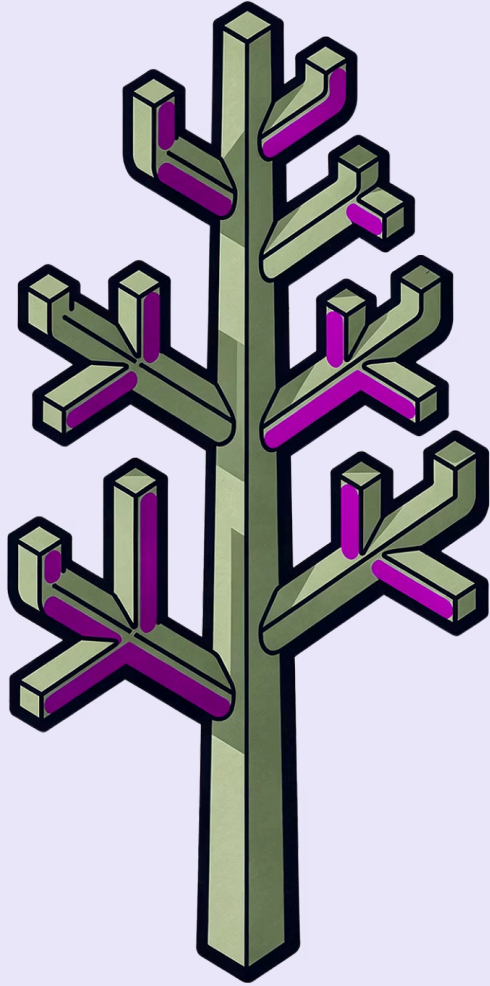




Decision Trees: Unpacking the Logic of Classification

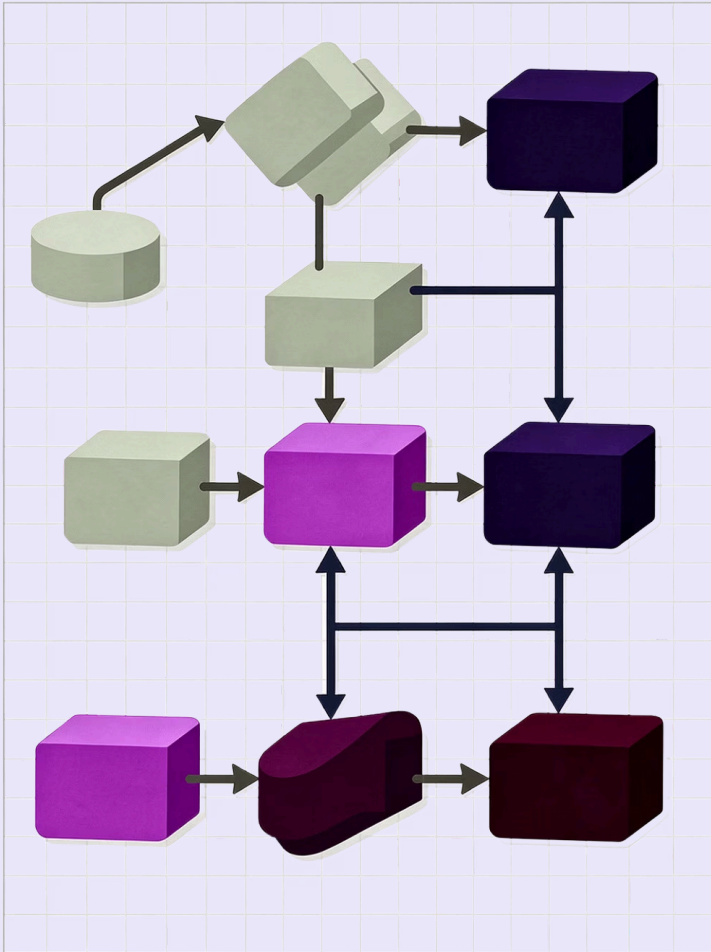
A deep dive into one of machine learning's most intuitive and interpretable algorithms — from the math behind every split to a real-world fruit classification example.



MACHINE LEARNING

SUPERVISED LEARNING

What is a Decision Tree?



Non-Parametric

A supervised learning method for both **classification** and **regression** tasks — no assumptions about data distribution required.

If-Then-Else Logic

Learns simple decision rules directly from data features, mimicking natural human reasoning and decision-making processes.

Flowchart Structure

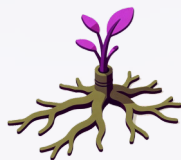
Visualized as a branching flowchart, making it one of the most **interpretable** and transparent models in machine learning.

Flexible Data Handling

Handles both **numerical** and **categorical** features with minimal data preprocessing required compared to other algorithms.

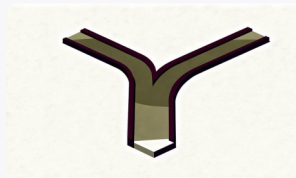
The Anatomy of a Decision Tree

Every decision tree is composed of four fundamental building blocks, each playing a distinct role in the classification process.



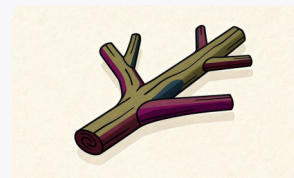
Root Node

The **starting point** of the tree, representing the entire dataset before any splits have been applied. The best feature is chosen here first.



Internal Nodes

Each internal node represents a **test on a feature** — for example, *"Is height > 170cm?"* — driving the data toward a classification.



Branches

The **outcome of each test** — typically "Yes" or "No" — that routes each data sample down a specific path through the tree.



Leaf Nodes

Terminal nodes at the bottom of the tree where no further splitting occurs — they hold the final class label or prediction output.

The Math Behind the Splits

The core objective of a decision tree is to find splits that produce the **purest possible child nodes** — nodes where all samples belong to the same class. This is quantified using impurity measures.

Information Gain (IG)

Measures the reduction in impurity achieved by splitting on a feature. The algorithm selects the feature and threshold that **maximizes IG**.

$$IG = \text{Impurity}(\text{Parent}) - \left[\frac{N_L}{N} \cdot \text{Impurity}(L) + \frac{N_R}{N} \cdot \text{Impurity}(R) \right]$$

Gini Impurity

Measures the probability of **misclassifying** a randomly chosen element. A Gini of 0 means a perfectly pure node.

$$\text{Gini}(p) = 1 - \sum_{i=1}^C p_i^2$$

Entropy

Measures the **level of disorder** or uncertainty in a dataset. Entropy of 0 indicates perfect purity; higher values indicate more mixing.

$$H(p) = - \sum_{i=1}^C p_i \log_2(p_i)$$

❗ In both measures, **p_i** is the proportion of samples belonging to class *i*, and **C** is the total number of classes. Lower impurity after a split = better split.

Real-World Example: Predicting Fruit Type

Let's classify fruits using just two features — **Color** and **Diameter** — to see how a decision tree builds its logic step by step.

Training Dataset

Color	Diameter (cm)	Fruit
Green	3	Apple
Red	3	Apple
Red	1	Grape
Red	1	Grape
Yellow	3	Lemon

Tree Decision Rules

Split 1 — Diameter

Diameter $\geq$ 3: {Apple, Apple, Lemon}

Diameter $<$ 3: {Grape, Grape} →
Leaf: Grape ✓

Split 2 — Color

Color = Yellow: {Lemon} →
Leaf: Lemon ✓

Color $\neq$ Yellow: {Apple, Apple} →
Leaf: Apple ✓

✔ Result: 100% accuracy on training data using just 2 splits — clean, interpretable, and efficient!

Advantages & Disadvantages

✓ Advantages

- **Interpretable:** Visualizable as a flowchart — anyone can follow the logic.
- **Low Prep:** Requires minimal data preprocessing — no normalization needed.
- **Flexible:** Handles both numerical and categorical features natively.
- **White Box:** Decision logic is fully transparent and easily explained.

⚠ Disadvantages

- **Overfitting:** Prone to creating overly complex trees that memorize noise in training data.
- **Instability:** Small variations in data can produce drastically different tree structures.
- **Piecewise Constant:** Predictions are not smooth — they jump between discrete values at boundaries.
- **NP-Complete:** Finding the truly optimal tree is computationally infeasible; algorithms rely on greedy heuristics.

Decision Trees: Building Blocks for Powerful Models

Decision trees are far more than a standalone classifier — they are the **foundational unit** of some of the most powerful ensemble algorithms in modern machine learning.



Single Decision Tree

Intuitive, interpretable, and a perfect introduction to ML concepts. Best for simple problems and explainability requirements.



Random Forest

An ensemble of hundreds of trees trained on random subsets of data and features — dramatically reduces overfitting and variance.



Gradient Boosting (XGBoost, LightGBM)

Trees built *sequentially*, each correcting the errors of the last. State-of-the-art accuracy on structured/tabular data competitions.

- ❏ Mastering the single decision tree — its splits, impurity measures, and structure — is the essential first step to understanding **all ensemble methods** in machine learning.